Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: Sampay Aquaculture Co-Pilot
b. Team leader name: Ilfa Hidayat
c. Problem Statement: Future of Work & Enterprise Productivity

Brief about the Idea

CORE CONCEPT

Agentic AI Operating System

Sampay Aquaculture Co-Pilot is an agentic AI operating system that transforms high-stakes, labor-intensive biofloc aquaculture into an efficient, intent-driven enterprise.

MARKET SCALE

1.56M+ Tons

Nile Tilapia Market

Anchored by Indonesia's massive Nile Tilapia (*Oreochromis niloticus*) market—the #1 freshwater commodity with annual production exceeding **1.56 million tons** (KKP & BPS data).

FUTURE OF WORK

Natural Language Ops (NLOps)

Farm workers are elevated from grueling manual calculations & monitoring to strategic supervisors.

Empowers teams to manage complex pond ecosystems seamlessly via plain language in Bahasa Indonesia or English.

Opportunities & Key Solution USPs

OPPORTUNITIES & PROBLEM SOLVING

Active Agentic Co-Pilot: Unlike generic sensor dashboards displaying raw numbers, Sampay acts as an active co-pilot combining RAG scientific reasoning (biofloc SOPs) with direct IoT hardware execution.

Automated OpEx & Risk Reduction: Automates biochemical calculations (C/N ratios, molasses dosing, daily feeding weights), directly eliminating human error driving 70% of crop failures and optimizing 60%–70% feed OpEx.

UNIQUE SELLING PROPOSITIONS (USP)

NLOps: Bridge physical IoT telemetry and human intent instantly through conversational commands in Bahasa Indonesia or English.

Zero-Hallucination Biofloc Reasoning: Grounded RAG knowledge base ensuring chemical and feeding recommendations strictly adhere to peer-reviewed science (*Avnimelech, MDPI 2025*).

Closed-Loop Enterprise Automation: Supervised Co-Pilot mode to autonomous Auto-Pilot edge execution via Gemini Function Calling.



List of Features Offered by the Solution

Proactive Daily Briefing

Automated morning synthesis of 24-hour sensor logs (DO, pH, Ammonia, Temp) delivering precise feeding schedules and safety scores.

Intent-Based Execution

Human-in-the-loop natural language command authorization for real-time, critical pond interventions.

Automated IoT Orchestration

Secure REST/MQTT control signals dispatching commands directly to edge actuators (aerators, auto-feeders, dosing valves).

Multi-Pond Enterprise Dashboard

Unified monitoring enabling a single supervisor to oversee multiple distributed biofloc ponds simultaneously.



Process flow diagram

01

Telemetry Ingestion

ESP32 sensors send periodic water quality logs to Node.js backend.

ESP32 Node.js

02

AI Synthesis & Reasoning

Google Gemini API analyzes telemetry against Biofloc RAG SOPs.

Gemini API RAG SOPs

03

Intent / Decision

Farm supervisor reviews automated briefing or inputs natural language command.

Human-in-Loop

04

Execution & Action

Gemini triggers tool calls saved in Cloud SQL & routes to ESP32 actuators.

Cloud SQL Knex.js

CLOSED-LOOP AUTOMATION ARCHITECTURE

This integrated pipeline continuously bridges telemetry collection, Gemini RAG intelligence, supervisor oversight, and edge physical actuation—creating an active, highly accurate co-pilot for biofloc smart farming.

Wireframes / Mock Diagram Solution

NLOps Natural Language Chat Interface

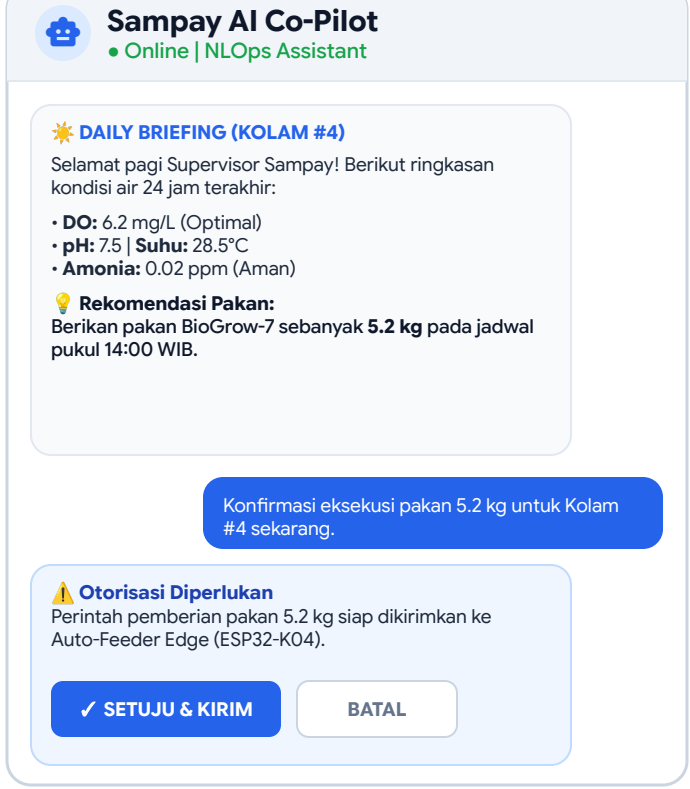
Antarmuka percakapan interaktif berbasis AI Co-Pilot yang menyajikan ringkasan kondisi kesehatan kolam secara proaktif setiap pagi (Daily Briefing) serta mengontrol ekseksi perangkat IoT melalui konfirmasi instruksi bahasa alami.

Human-in-the-Loop Execution Control

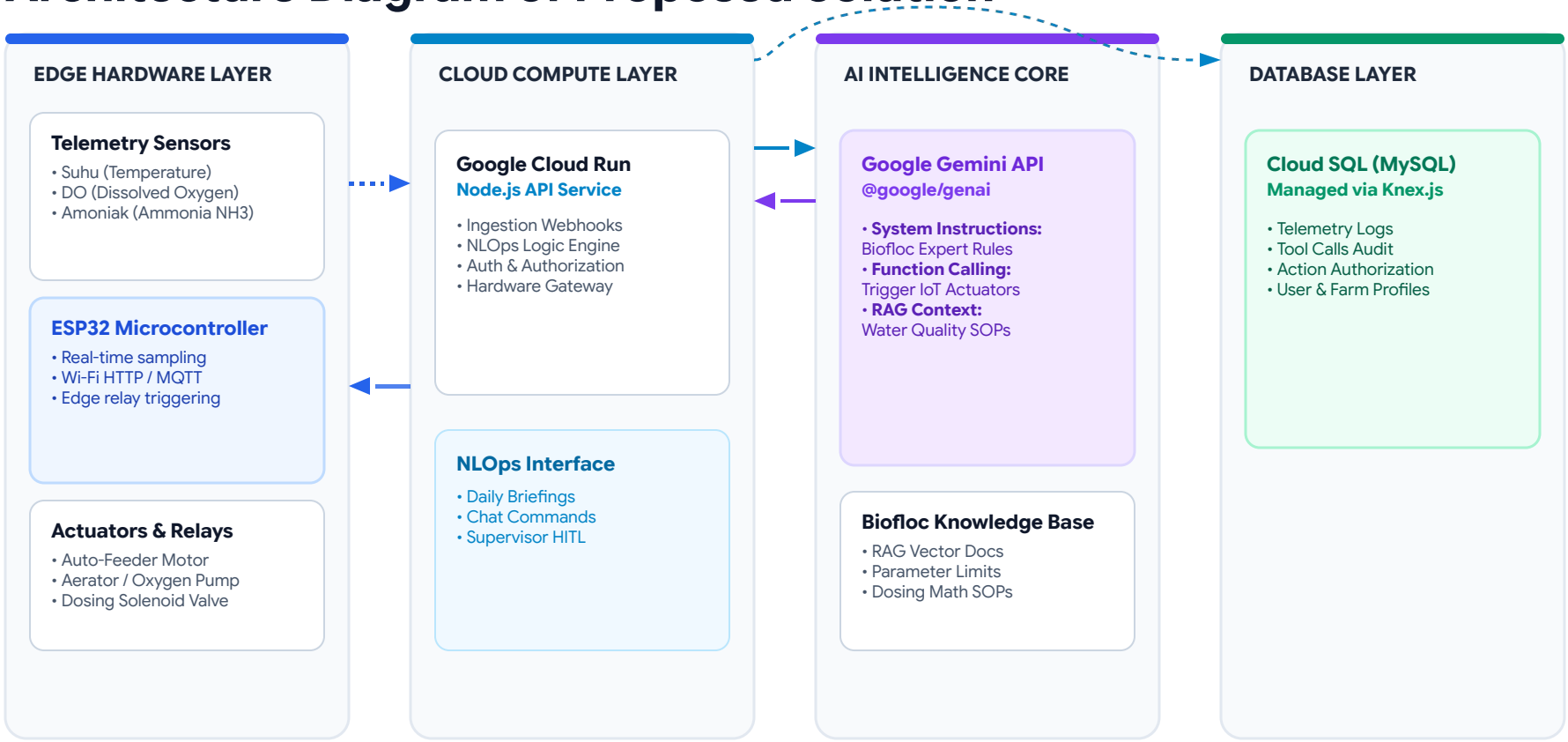
Menjamin keamanan operasional tambak biofloc dengan memberlakukan otorisasi manual oleh supervisor sebelum sinyal aksi (pemberian pakan, aerasi, atau obat) dikirimkan secara langsung ke aktuator edge hardware.

HIGHLIGHT FITUR UTAMA

- Sintesis otomatis data sensor 24 jam (DO, pH, Suhu, Amonia)
- Rekomendasi dosis pakan presisi berbasis RAG SOP Biofloc
- Tombol aksi cepat eksekusi otomatis tanpa perlu input manual yang rumit



Architecture Diagram of Proposed Solution



Technologies to be used in the solution

AI & LLM SDK

Gemini API
@google/genai

- **Core Reasoning:** Advanced contextual processing via Gemini latest models.
- **Function Calling:** Structured execution triggers directly interfacing with edge hardware relays.
- **RAG Integration:** Augmented retrieval from custom Biofloc expert knowledge base SOPs.

BACKEND & CLOUD

Google Cloud Run
Node.js Service

- **Serverless Compute:** Auto-scaling container runtime handling all active webhooks.
- **NLOps Logic Engine:** Processes natural language commands into actionable payloads.
- **Secure Gateway:** Authentication and encrypted routing to hardware microcontrollers.

DATABASE & ORM

Cloud SQL (MySQL)
Knex.js Query Builder

- **Managed Persistence:** High-availability SQL instances for continuous operational data.
- **Telemetry Logging:** Stores real-time sensor metrics (pH, DO, Suhu, Amonia).
- **Audit & Authorization:** Logs supervisor approvals and complete AI tool execution history.

IOT HARDWARE & EDGE

ESP32 Controllers
MQTT Protocol Focus

- **MQTT Messaging:** Low-latency lightweight publish/subscribe protocol for IoT telemetry.
- **Edge Sampling:** Real-time data acquisition from physical water quality sensors.
- **Actuator Control:** Relay triggering for feeder motors, aerators, and dosing valves.

Estimated Implementation Cost (BOM & OpEx)

1. HARDWARE BOM (PER POND)

Industrial Biofloc Node
Per-Pond Automation Hardware

- **ESP32 MCU + RS485 Modbus**
  ~$6.00 USD
- **VFD (0.75kW/1HP) Motor Driver**
  ~$45.00 USD (30-40% OpEx savings)
- **Industrial Analog DO Sensor Kit**
  ~$160.00 USD (Hypoxia protection)
- **Multi-Parameter (pH, TDS, Temp)**
  ~$18.00 USD
- **Smart Auto-Feeder Mechanism**
  ~$35.00 USD
- **Industrial Panel, Power & Relays**
  ~$11.00 USD

TOTAL COMMERCIAL HARDWARE / POND
~$275.00 USD
~Rp 4.9M IDR

2. CLOUD INFRASTRUCTURE (OPEX)

Google Cloud Serverless
Monthly Operating Expenditure

- **Google Cloud Run (Node.js API)**
  $0.00 / month (Fits in Free Tier)
- **Cloud SQL (MySQL Instance)**
  ~$10.00 – $15.00 / month
- **Google Gemini API (@google/genai)**
  Pay-as-you-go (~$0.002 / 1k tokens)

ESTIMATED CLOUD OPEX
~$15 - $20 USD
per month (shared system)

3. ECONOMIC FEASIBILITY

Scalability & Impact
Smallholder Farmer Accessibility

Key Takeaway:

Accessible commercial entry barrier (**$275/pond** full automation node), making **Sampay** highly viable and scalable for tilapia farms across Indonesia.

By leveraging Google Cloud's free tier and pay-as-you-go AI models, operational cloud costs remain minimal.

Snapshots of the prototype

Prototype Performance report/Benchmarking

Additional Details/Future Development (if any)

Provide links to your:

- GitHub Public Repository
- Demo Video Link (3 Minutes)
- Final Product Link

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

